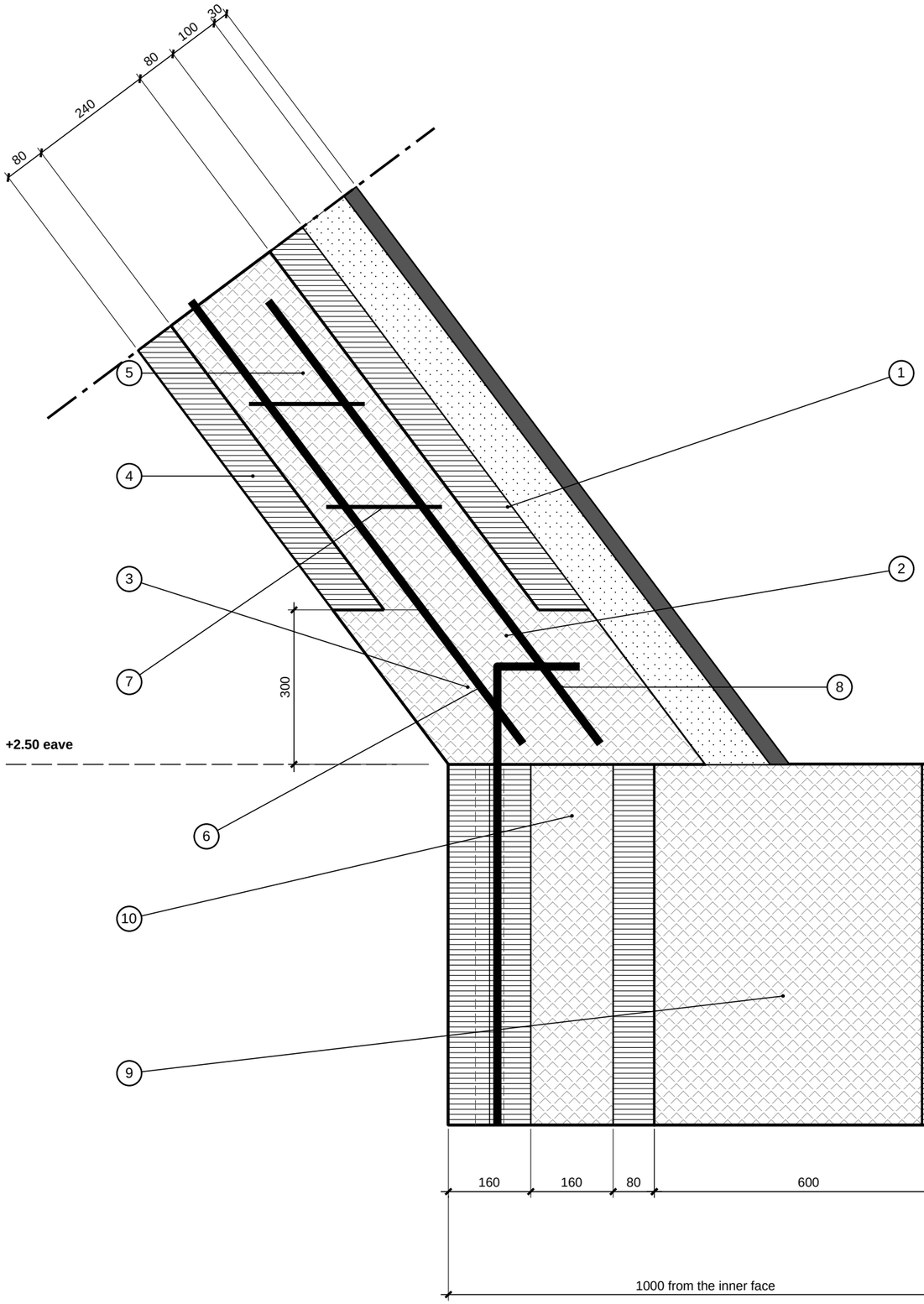
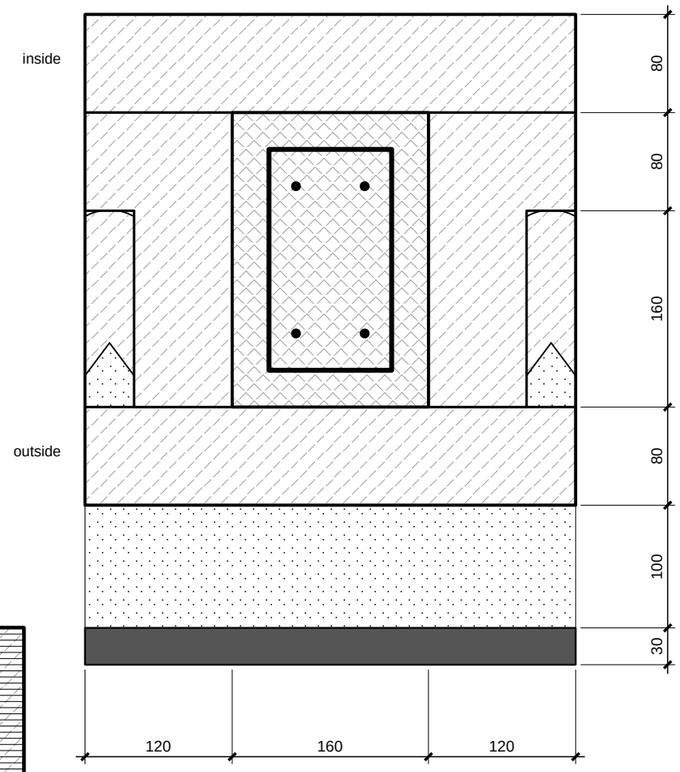


Plan cut through a rib at one layer, 1:5



Detail D1: eave, roof to wall

1. Roof outer layer 80, printed with the rest of the piece; outside it the insulation 100 (dotted) and the outer layer 30 (dark).
2. Zig-zag lattice 80 in a 160 zone; it touches the outer layer and layer 2 every @ 600; the cells between are open air pockets for ventilation.
3. Layer 2, 80: printed bead; beside each rib it turns into the printed channel walls (see the plan cut).
4. Inner layer 1, 80, room side. Layers 1 and 2 together are the structural leaf: 160 at the base, 80 at the crown (layer 2 ends).
5. Rib (cast column): channel 160 wide x 240 deep through layer 2 and the 160 zone, filled with poured concrete; section cut along the rib. It rises out of the solid base in the same pour, without a joint.
6. Rib bars R, 4 Ø16 (2 in the cut), anchored in the solid base where they cross the hooks of the wall bars A (assumed; steel fibre in the concrete could replace them).
7. Stirrups Ø8 every 200 (assumed).
8. Solid base of the roof: the first 300 (vertical) above the eave is full cast concrete over the whole 400, no air zone, poured in one go with the rib above it; wall bars A hook into it. Replaces the ring beam.
9. Buttress 400 wide, 1000 from the inner face (600 beyond the wall): printed formwork (80 end cap shown) filled with cast concrete; top flush with the roof base at +2.50.
10. Wall 400 at the buttress: printed inner layers and outer layer, the 160 zone between them cast and one body with the buttress (no insulation here, the cut is through the buttress, not the curtain wall). Wall bars A Ø16 run from the footing up into the solid base and end in a 160 hook (assumed). The roof grid is A or B (see the roof-grid sheets); the detail is cut along a rib.

0 1 2 3 4 dm

Pyramid roof · 3D-printed concrete · GSAPP
Scale 1:10 at A3 · rib plan cut 1:5 · dimensions in mm
2026-10-14 · hand estimates, not engineered